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$$= \frac{1}{\frac{x^2+y^2}{x^2}} \cdot \frac{1}{x}$$

$$= \frac{x^2}{x^2+y^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2}$$

partially diff. w.r.t. 'y' again

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial \left(\frac{x}{x^2+y^2}\right)}{\partial y^2}$$

$$= x \frac{\partial (x^2+y^2)^{-1}}{\partial y^2}$$

$$= x \cdot (-1)(x^2+y^2)^{-2} \frac{\partial (x^2+y^2)}{\partial y}$$

$$= -x(x^2+y^2)^{-2}(2y)$$

$$= \frac{-2xy}{(x^2+y^2)^2}$$

Now, Diff. w.r.t. 'x'

$$\frac{\partial z}{\partial x} = \frac{\partial (\tan^{-1}(y/x))}{\partial x}$$

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{\partial (y/x)}{\partial x}$$

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11. If $u = e^{ax+by}\phi(ax-by)$ prove that

$$b\frac{\partial u}{\partial x} + a\frac{\partial u}{\partial y} = 2abu.$$

sol^n:-

$$u = e^{ax+by}\phi(ax-by)$$

diff. w.r.t. 'x' partially

$$\frac{\partial u}{\partial x} = \frac{\partial\left(e^{ax+by}\phi(ax-by)\right)}{\partial x}$$

$$= e^{ax+by}\frac{\partial(\phi(ax-by))}{\partial x} + \phi(ax-by)\frac{\partial(e^{ax+by})}{\partial x}$$

$$= e^{ax+by}\phi'(ax-by)\frac{\partial(ax-by)}{\partial x} + \phi(ax-by)\cdot e^{ax+by}\cdot\frac{\partial(ax+by)}{\partial x}$$

$$= e^{ax+by}\phi'(ax-by)(a) + \phi(ax-by)e^{ax+by}\cdot(a)$$

$$\frac{\partial u}{\partial x} = e^{ax+by}\left[a\phi'(ax-by) + a\phi(ax-by)\right] \quad \text{—(i)}$$

Now,
partially diff. w.r.t. 'y'

$$\frac{\partial u}{\partial y} = \frac{\partial\left[e^{ax+by}\phi(ax-by)\right]}{\partial y}$$

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$$= abe^{ax+by}\phi'(ax+by) + abe^{ax+by}\phi(ax-by)$$

$$- abe^{ax+by}\phi'(ax-by) + abe^{ax+by}\phi(ax-by)$$

$$= 2abe^{ax+by}\phi(ax-by)$$

$$\left[\because e^{ax+by}\phi(ax-by) = u\right]$$

$$= 2abu$$

Here L.H.S. = R.H.S.

Hence proved.

12. Let $f = x^2\sin(y/x)$

solve:-

diff. partially w.r.t. 'x'

$$\frac{\partial f}{\partial x} = \frac{\partial(x^2\sin(y/x))}{\partial x}$$

$$= x^2\frac{\partial(\sin(y/x))}{\partial x} + \sin(y/x)\frac{\partial(x^2)}{\partial x}$$

$$= x^2\cos(y/x)\frac{\partial(y/x)}{\partial x} + \sin(y/x)(2x)$$

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13. If $u = x^y$ show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$

sol^n:-

$$\frac{\partial u}{\partial x} = \frac{\partial (x^y)}{\partial x} = y x^{y-1}$$

$$\frac{\partial u}{\partial y} = \frac{\partial (x^y)}{\partial y} = x^y \log(x)$$

$$\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}\left(\log(x) \cdot x^y\right)\right)$$

$$= \frac{\partial}{\partial x}\left[\log x \frac{\partial (x^y)}{\partial x} + x^y \cdot \frac{\partial (\log x)}{\partial x}\right]$$

$$= \frac{\partial}{\partial x}\left[\log x \cdot y x^{y-1} + x^y \cdot \frac{1}{x}\right]$$

$$= \frac{\partial}{\partial x}\left[\log(x) \cdot y x^{y-1} + x^{y-1}\right]$$

$$\log(x) \frac{\partial (y x^{y-1})}{\partial x} + y x^{y-1} \frac{\partial (\log(x))}{\partial x}$$

$$+ \frac{\partial (x^{y-1})}{\partial x}$$

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$$= \frac{\partial}{\partial x}\left(x^{y-1} + y \cdot x^{y-1}\log(x)\right)$$

$$= (y-1)x^{y-2} + y\left\{\frac{\partial}{\partial x}\left(yx^{y-1}\log(x)\right)\right.$$

$$= (y-1)x^{y-2} + y\left[x^{y-1}\cdot\frac{1}{x} + \log x \cdot (y-1)x^{y-2}\right]$$

$$= (y-1)(x)^{y-2} + yx^{y-2} + y^2\log x\,(y-1)x^{y-2}$$

$$= x^{y-2}\left[(y-1) + y + y^2\log x\,(y-1)\right]$$

$$= x^{y-2}\left[\log(x)(y-1)y^2 + y + (y-1)\right]$$

hence proved.

$$\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$$

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$$= t^n e^{\frac{-r^2}{4t}} \cdot \frac{1}{4t} \cdot \frac{\partial(-r^2)}{\partial r}$$

$$= t^n e^{\frac{-r^2}{4t}} \cdot \frac{1}{\cancel{4}_{2}t} (-\cancel{2}r)$$

$$= \frac{-r}{2} t^n \cdot e^{\frac{-r^2}{4t}} \cdot \frac{1}{t}$$

$$= \frac{-r}{2} t^{n-1} e^{\frac{-r^2}{4t}} \qquad \left[t^n \frac{1}{t} = t^n t^{-1} = t^{n-1} \right]$$

$$= \frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial}{\partial r}\left(\frac{-r^3}{2} t^{n-1} e^{\frac{-r^2}{4t}}\right)$$

$$= \frac{-t^{n-1}}{2}\left[r^3 \frac{\partial}{\partial r}\left(e^{\frac{-r^2}{4t}}\right) + e^{\frac{-r^2}{4t}} \frac{\partial}{\partial r}(r^3) \right]$$

$$= \frac{-t^{n-1}}{2}\left[r^3 e^{\frac{-r^2}{4t}} \frac{\partial}{\partial r}\left(\frac{-r^2}{4t}\right) + 3r^2 e^{\frac{-r^2}{4t}} \right]$$

$$= \frac{-t^{n-1}}{2}\left[r^3 e^{\frac{-r^2}{4t}} \left(\frac{-2r}{4t}\right) + 3r^2 e^{\frac{-r^2}{4t}} \right]$$

$$= \frac{-t^{n-1}}{2} e^{\frac{-r^2}{4t}} \left[r^3 \left(\frac{-2r}{4t}\right) + 3r^2 \right]$$

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$$= \frac{t^{n-1}}{2} e^{-\frac{r^2}{4t}} \left[\frac{r^2}{2t} + 3 \right] = \frac{t^{n-1}}{2} e^{-\frac{r^2}{4t}} \left[\frac{r^2}{2t} + 2n \right]$$

$$\frac{r^2}{2t} + 3 = \frac{r^2}{2t} + 2n$$

$$n = \frac{-3}{2} \text{ Ans.}$$

16. If $v = r^m$ and $r = \sqrt{x^2+y^2+z^2}$ show that—

$$V_{xx} + V_{yy} + V_{zz} = m(m+1) r^{m-2}.$$

soln :—

$$v = r^m, \quad r = \sqrt{x^2+y^2+z^2}$$

let $x = r\cos\theta$ and $y = r\sin\theta$

$$x^2+y^2 = r^2$$

$$r = \sqrt{x^2+y^2}$$

$$\frac{\partial r}{\partial x} = \frac{\partial (x^2+y^2)^{1/2}}{\partial x}$$

$$= \frac{1}{2}(x^2+y^2)^{-1/2} \frac{\partial (x^2+y^2)}{\partial x}$$

$$= \frac{1}{2}(x^2+y^2)^{-1/2} (2x)$$